

LUMBINIX 2026

Sākṣī

A witness network that turns everyday visitor photos into evidence a conservator can trust.

- Live today at project-saksi.vercel.app, in English and Nepali
- Piloted at Lumbini, the birthplace of the Buddha
- Built by the LumbiniX Committee

The problem

Heritage sites change quietly. Nobody is watching closely enough to notice in time.

Once every few years

how often a full condition survey happens

Thousands

of visitors pass through every single week

Zero

of those visits currently produce usable evidence

The people are already there. The system to use what they see is not.

What that gap actually costs

Manga Hiti is a stone water spout in the Kathmandu Valley, carved in the sixth century.

- A 2019 valley wide survey found fewer than half of these spouts still flowing.
- Almost none of the ones still standing are checked on any regular schedule.
- By the time decline gets noticed, it is often already advanced.

<50%

of valley stone spouts still flow, per the 2019 survey

A tourist photo cannot fix this

1

Position

The same spot, every time, or nothing lines up.

2

Honesty

Measured, or just a rough guess by eye. Most photos do not say.

3

Record

Who took it, when, and how it was checked.

Without all three, more photos just means a bigger pile nobody can use.

Our idea, in one picture

Then

- A historical photo of the exact spot.

Now

- A fresh photo, aligned to match it exactly.

Visitors fade between the two, and the app builds a dated series over time. That series is the whole product.

WHAT WE BUILT

The product

Three parts. One idea: turn a visit into evidence.

Three parts, one app

1

Tīrtha

The place. Explore a site and see how it has changed.

2

Sākṣī

The witness. Capture a photo that counts as real evidence.

3

Dhamma

The knowledge. Ask a question, get an answer with a real source.

Three destinations. No feed, no clutter, nothing to scroll for its own sake.

Tīrtha: explore and compare

- An interactive map of the site, with a friendly guide you can ask anything.
- Every historical photo carries an honest label for how much you can trust it.
- Quests send a visitor to the viewpoint that has gone longest without a new photo, not the most popular one.

Sākṣī: the witness loop

1

Pick a viewpoint

Choose a fixed vantage point already mapped for the site.

2

Line it up

The phone scores your position, direction and tilt against the target.

3

Capture

The shutter unlocks only once the match is genuinely good.

4

Record

The photo, its score, and the moment are saved, ready to compare later.

Facing the right direction matters most. You cannot lock the shot while facing the wrong way.

Honesty is the core idea

Measured

- Position and heading confirmed against the target.
- Stored as a trustworthy, comparable record.

By eye

- No lock achieved, captured anyway.
- Clearly labelled as an estimate, never dressed up as measured.

Anyone reading the data later can filter by trust level. That is what makes the whole record usable.

Real AI, running on the phone

A crack detection model runs on the device itself, fully offline.

- It finds candidate cracks the moment a photo is captured.
- The AI only suggests. A person always decides how serious it is.
- Every AI-assisted report is flagged as such in the record.

82%

honest detection accuracy, shown in the app, never rounded up

Closing the loop

Every report a visitor files reaches the people responsible for that site.

- A custodian can acknowledge, mark in progress, or resolve a report, with a note.
- Access is invite only. Each custodian sees only their own site, enforced by the database itself.
- Reports export to CSV and GeoJSON, ready for a real GIS workflow.

Live

today at project-saksi.vercel.app/custodian

Dhamma: cite it, or refuse

Ask about Buddhist teaching or heritage conservation.

- Every answer is built only from real, checkable sources, with a citation.
- When the sources do not support an answer, it says so instead of guessing.
- Every sentence in a cached answer is checked against the source it names, not just that the source exists.

74 / 74

correct on our benchmark, zero unsupported answers

Recognition without gaming the system

- Contribution is recognized, but merit cannot be spent, traded, or gamed like points.
- The Guardians leaderboard is built only from real evidence uploaded, calculated on the server.
- Nobody can inflate their standing without actually doing the work.

SĀKṢĪ

Beyond Lumbini: the platform

Three more UNESCO sites in the Kathmandu Valley now run on the exact same app.

- Patan Durbar Square, Changu Narayan, and Manga Hiti. Same rules, same discipline, no rewrite.
- Lumbini was the pilot. The Valley sites are proof it generalises.

1,200+

UNESCO World Heritage sites worldwide face the same gap

HOW IT WORKS, HOW IT LASTS

Technology & business

What is under the hood, and why it can keep running.

Architecture, at a glance

On the phone

Every record saves locally first. The heavy AI work, crack detection and offline answers, runs on the device.

↓

In the cloud

Supabase holds a protected copy. Row level security means each person only sees what they are allowed to.

↓

On the web

A public bilingual site and a live custodian dashboard: the front door for visitors and institutions alike.

185 automated tests check this behaviour every time the code changes.

Tools we used

ON THE DEVICE

- React Native & Expo
- TypeScript
- SQLite
- onnxruntime, for the crack detector
- llama.rn, for offline answers
- MapLibre, for the map

CLOUD & WEB

- Supabase, database and sign in
- Next.js, the public site
- Ollama Cloud, for online answers

How this can last

Free for every visitor, always.

- Revenue comes from licensing the custodian dashboard to heritage authorities, and from conservation grants.
- Running costs stay low: on-device inference, static retrieval, open map tiles.
- No money is ever taken from a business operating inside a monitored site. That rule is written into a public ethics policy.

0

sponsor money accepted from inside a monitored site

Where things stand today

Built past the hackathon, not stopped at it.

185

automated tests passing, app and web dashboard combined

74 / 74

on the Dhamma knowledge benchmark, zero fabricated citations

2

languages live on the public site: English and Nepali

4

UNESCO sites covered: Lumbini, plus three in the Kathmandu Valley

What comes next

PHASE ONE, THIS YEAR

Prove it with one partner

Run a full monitoring cycle with a real Nepali heritage authority against the live dashboard.

PHASE TWO, ONE TO THREE YEARS

Become a platform for Nepal

Cover more of Nepal's protected monuments, and publish the app on the Play Store.

PHASE THREE, THREE PLUS YEARS

The same gap, everywhere

Nothing in the design is specific to Nepal. Any of the 1,200+ UNESCO sites can use it next.

What we have not solved yet

- Some site coordinates come from old documents, not a fresh survey, so they are estimates.
- The crack detector only finds cracks, at 82 percent accuracy. It is a helper, not a verdict.
- A few Kathmandu Valley sites do not yet have a before and after photo pair.
- No heritage authority has signed on as a partner yet. Outreach is underway.
- The Android build is ready. Publishing it to the Play Store is still in progress.

The team

Built by the LumbiniX Committee.

AS

Aaditya Sapkota

SS

Siddanta Sodari

Alongside the wider LumbiniX Committee team.



THANK YOU

Sākṣī

A witness that guides, an AI that does not lie, a custodian who can act, and a record worth trusting.

- Live today at project-saksi.vercel.app, in English and Nepali
- Built by the people already standing in front of the monument
- Questions welcome